Course outline

1. Linear ordinary differential equations (ODEs) and initial value problems (IVPs)
2. Power series solutions to ODEs
3. Laplace transforms
4. Fourier series
5. Boundary value problems (BVPs) and linear partial differential equations (PDEs)
6. Fourier transforms

Topic 3: Laplace transforms

DE in t $\to$ n (discrete var)
$\to$ s (cts var)

How Topic 3 builds on Topic 2:

- Functions in, and solutions to, the governing ODEs in Topics 1 and 2 are generally continuous – or at least, we focus on intervals where these functions are continuous.
- In Topic 3, we tackle a new class of ODEs where discontinuous and impulsive functions are involved.

Topic 3: Laplace transforms

Leaning outcomes:

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1. Understand how ODEs involving discontinuous and impulsive functions can arise in modelling of real-world problems
2. Know how to calculate Laplace transforms and inverse Laplace transforms of appropriate functions
3. Know when and how to use the Laplace transform method for solving ODEs and IVPs

Lecture 10

We will:

1. Revise definition and calculation of improper integrals
2. Learn to use comparison test to determine convergence of an integral
3. Learn the definition of the Laplace transform and its basic properties
4. Discuss application of the Laplace transform to piecewise continuous functions

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2. or f is not bounded on (a, b).

- Assume all unboundedness occurs at a and/or b (otherwise split up the integral).
- The improper integral is then defined as

$$\int_a^b f(t)\mathrm{d}t = \lim_{q \to a^+} \lim_{r \to b^-} \int_q^r f(t)\mathrm{d}t$$

Exercise 3.1: Evaluate the following improper integrals

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$$= \lim_{r\to 0^-}\left[-\frac{1}{t}\right]_{-1}^r + \lim_{q\to 0^+}\left[-\frac{1}{t}\right]_q^1 \ldots \text{does not exist}$$

Comparison test

Definition (Comparison test)

Let f and g be two continuous functions on (a, b).

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Exercise 3.2: Does $\displaystyle\int_0^\infty \underbrace{\{\sin(t)\,\mathrm{e}^{-t}\}}_{f(t)}\,\mathrm{d}t$ converge?

$|\sin t| \leq 1$

$g(t) = e^{-t}$

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$$\int_0^\infty \mathrm{e}^{-t}\,\mathrm{d}t \quad \text{converges (Ex 3.1a)}$$

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$$\left.\begin{array}{l} |\sin(t)\,\mathrm{e}^{-t}| \leqslant \mathrm{e}^{-t} \quad \text{for} \quad t \in (0, \infty) \\ \displaystyle\int_0^\infty \mathrm{e}^{-t}\,\mathrm{d}t \quad \text{converges (Ex 3.1a)} \end{array}\right\} \quad \int_0^\infty \{\sin(t)\,\mathrm{e}^{-t}\}\,\mathrm{d}t \quad \text{converges}$$

The Laplace transform

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The *Laplace transform*, $\mathscr{L}$, of the function $f(t)$ is

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Since it is a *linear transformation*:

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2. $\mathscr{L}\{\beta f\} = \beta\mathscr{L}\{f\} = \beta F(s)$ where β is a constant

Exercise 3.3: Find the Laplace transform of $f(t) = 1$

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$$= \lim_{a\to\infty} \left[-\frac{1}{s} \mathrm{e}^{-s\,t} \right]_{\cancel{0}\; t=0}^{a}$$

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$$= \lim_{a\to\infty} \left[-\frac{1}{s} \mathrm{e}^{-s\,t} \right]_0^a = -\frac{1}{s} \lim_{a\to\infty} \left[\cancelto{0}{\mathrm{e}^{-s\,a}} - 1 \right] = \frac{1}{s}$$

assuming $s > 0$

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Thus $\mathcal{L}\{1\} = F(s) = 1/s$

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The Laplace transform exists (integral converges) if we restrict to $\mathrm{Re}(s) > 0$.

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$$\mathcal{L}\{\mathrm{e}^{-t}\} = \int_0^\infty \mathrm{e}^{-s\,t}\,\mathrm{e}^t\;\mathrm{d}t = \lim_{a\to\infty}\int_0^a \mathrm{e}^{(1-s)\,t}\mathrm{d}t$$

Exercise 3.4: Find the Laplace transform of $f(t) = \mathrm{e}^t$

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$$= \lim_{a\to\infty}\left[\frac{\mathrm{e}^{(1-s)\,t}}{1-s}\right]_0^a = \frac{1}{s-1}$$

$$\left\| \frac{1}{1-s}\left(\cancelto{0}{e^{(1-s)t}} - 1\right) = \frac{1}{s-1}\right.$$

Thus $\mathcal{L}\{\cancel{f}\} = F(s) = \cancel{1/(1-s)}$ for $\mathrm{Re}(s) > 1$.

e^t

$1/s-1$

Existence of Laplace transform / Convergence of improper integral

Criteria to ensure the Laplace transform exists (improper integral converges)

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Piecewise continuous

- f has a finite number of discontinuities on $(0, L)$.
- The limits of f at the end points, and on each side of every discontinuity, are finite.

Existence of Laplace transform / Convergence of improper integral

Criteria to ensure the Laplace transform exists (improper integral converges)

- f is piecewise continuous on $[0, L]$ for all $L > 0$
- $|f(t)| \leqslant k\,\mathrm{e}^{\alpha\,t}$ for all $t > T$ for $\alpha, k, T \in \mathbb{R}$
  - i.e., f bounded by an exponential function
  - Laplace transform guaranteed to converge for $\mathrm{Re}(s) > \alpha$

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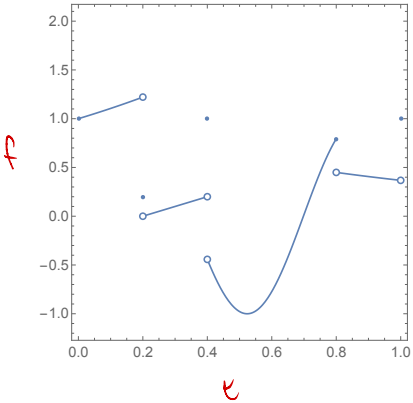
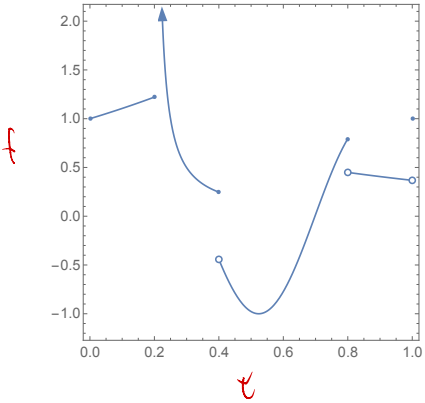
- f has a finite number of discontinuities on $(0, L)$.
- The limits of f at the end points, and on each side of every discontinuity, are finite.

Exercise 3.5: Are the following functions piecewise continuous?